NETWORKWALKS

PENETRATION TESTING REPORT

Footprinting and Network Scanning | Week 02 | B084

Report detail	My information
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Role and batch	Cybersecurity and Penetration Testing Intern | B084
Report date	11 November 2026
Practical activity date	10 November 2026, as shown in the dated outputs
Modules covered	W2-PM1 Multiple Kali Tools W2-PM5 Zenmap Scanning
Targets	networkwalks.com My own local network for the scanning activity
Authorization	Written permission secured for Networkwalks, as recorded in my original report
Phases covered	Reconnaissance and footprinting Network scanning and discovery

This report documents my Week 2 learning experience at Networkwalks. I used six Kali Linux tools to gather information about networkwalks.com and recorded what each result taught me. My screenshots are placed with the relevant tool sections. The results helped me understand how domain, website and DNS information can support a security assessment.

1 Liability Disclaimer

I carried out these learning activities on systems for which I had permission or on my own devices and network. I understand that this knowledge must be used responsibly and within the approved scope. I take responsibility for my actions and will not use these tools to access systems without authorization. These activities are for education and research, and Networkwalks, my instructor and the authors are not responsible for any misuse.

2 Introduction

During Week 2 of my Networkwalks internship, I worked on footprinting and reconnaissance using Kali Linux. I used WHOIS, WhatWeb, nslookup, curl, wafw00f and DNSRecon to collect different pieces of information about networkwalks.com. I learned how to read the outputs and explain what they mean without treating every visible detail as a security weakness.

I also used Zenmap for local network discovery. My scan identified 10 responding hosts, and I reviewed the topology view. I have included both screenshots with only four selected IP addresses visible and other local identifiers obscured.

3 Tools Used

Tool	How I used it
Kali Linux in VirtualBox	Ran my footprinting commands in the lab environment.
WHOIS	Reviewed public registration details and name servers.
WhatWeb	Identified website technologies and reported versions.
nslookup	Resolved the domain to its IP address.
curl -I	Inspected the website response headers.
wafw00f	Checked for a detectable web application firewall.
DNSRecon	Enumerated DNS records and related infrastructure.
Windows CMD and Zenmap	Tools for local network configuration and host discovery in W2-PM5.

4 Activities Performed

4.1 Footprinting and Reconnaissance

I used each tool for a specific purpose, then compared the results to understand the target better. The following pages contain my commands, screenshots, observations and lessons learned.

4.1.1 WHOIS Domain Information

```
whois networkwalks.com
```

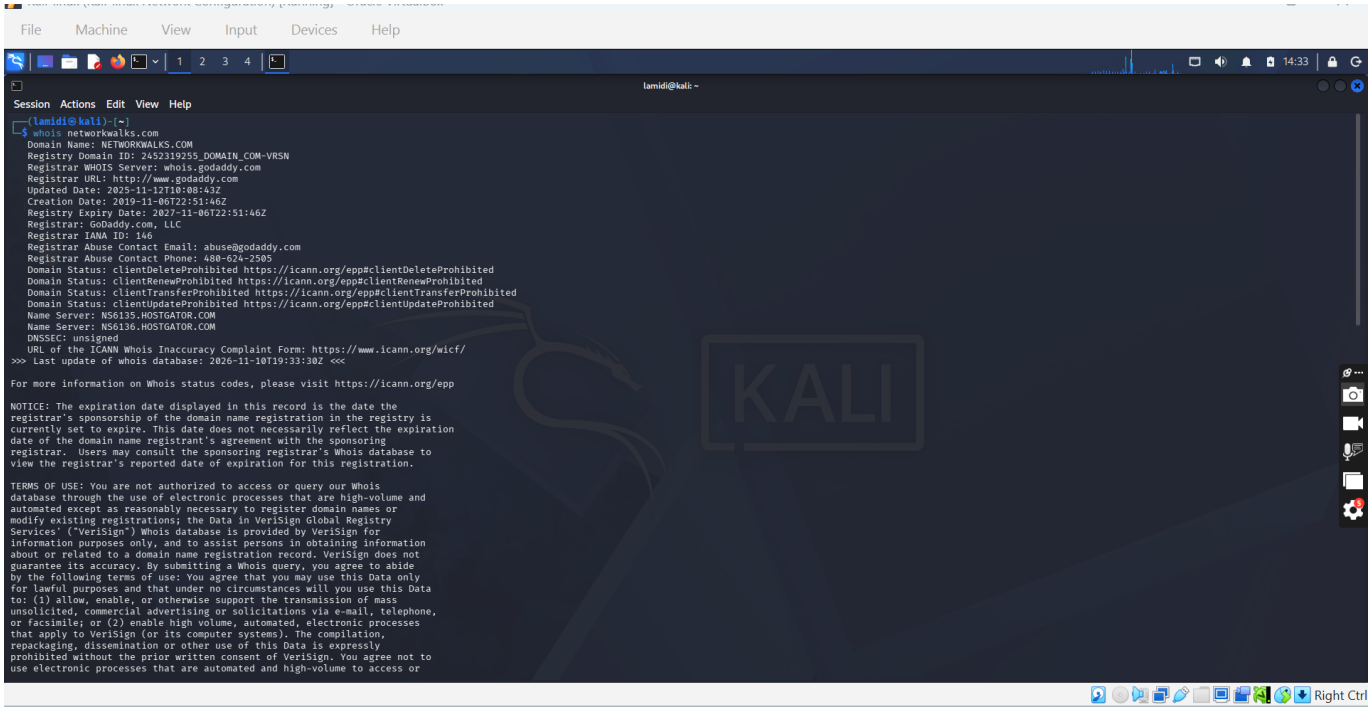


Figure 1 WHOIS output for networkwalks.com | Evidence E02

I started with WHOIS to learn about the domain registration. The output showed GoDaddy.com, LLC as the registrar, a creation date of 6 November 2019 and a registry expiry date of 6 November 2027. The listed name servers were NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM. DNSSEC was shown as unsigned.

This helped me understand how domain registration information provides a starting point for reconnaissance. The unsigned DNSSEC status is an observation for further review; it does not prove that the domain has been attacked or compromised.

4.1.2 WhatWeb Website Technologies

```
whatweb networkwalks.com
```

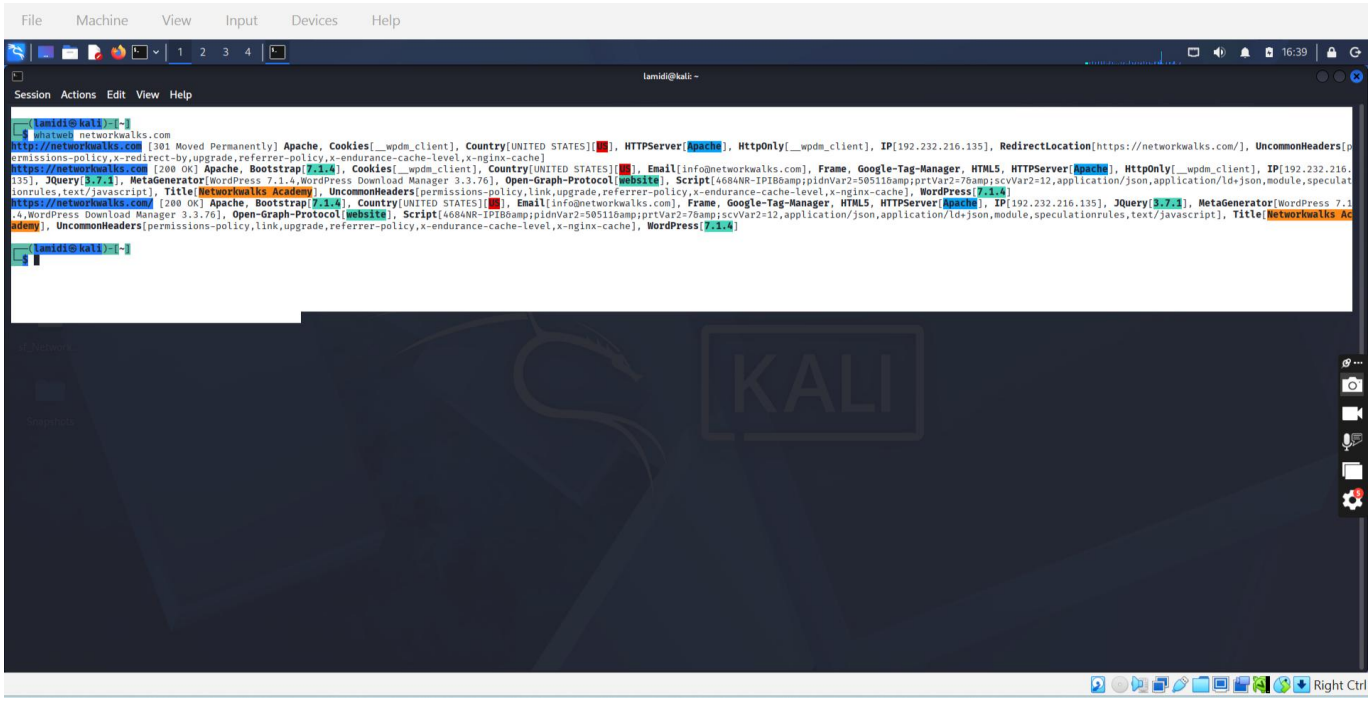


Figure 2 WhatWeb technology fingerprinting output | Evidence E03

I used WhatWeb to identify technologies visible on the website. The output showed an HTTP 301 redirect to HTTPS, followed by successful 200 OK responses. It identified Apache, WordPress 7.1.4, jQuery 3.7.1 and WordPress Download Manager 3.3.76. It also displayed the title Networkwalks Academy, the address 192.232.216.135 and the public email info@networkwalks.com.

I learned that website responses can reveal information about the software in use. These are the versions reported by WhatWeb, so I would verify them before comparing the software with security advisories. A reported version alone does not confirm a vulnerability.

4.1.3 nslookup Domain Resolution

```
nslookup networkwalks.com
```

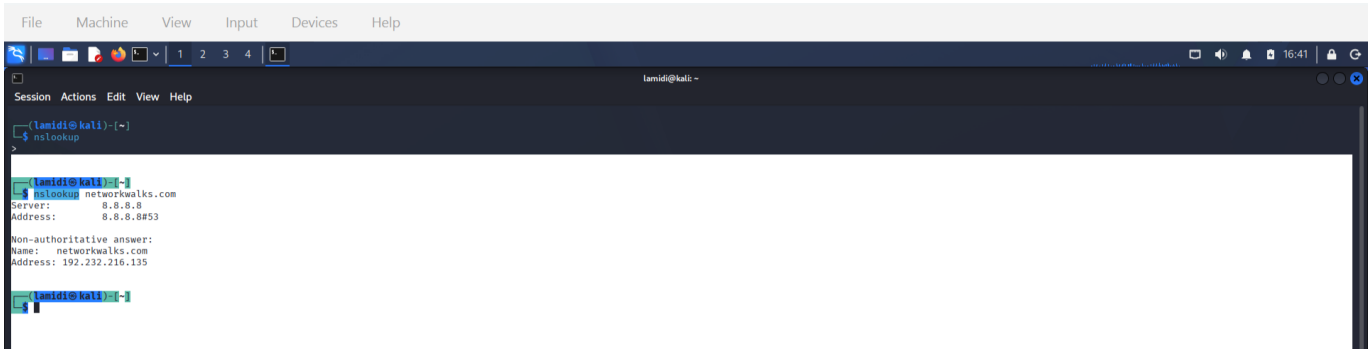


Figure 3 nslookup resolving the target domain | Evidence E04

I used nslookup to find the IP address associated with networkwalks.com. The query used the DNS server 8.8.8.8 on port 53 and returned 192.232.216.135 as a non-authoritative answer. This matched the address shown in my WhatWeb results.

I learned that a non-authoritative answer comes from the DNS resolver rather than directly from the domain's authoritative name server. It does not automatically mean the result is wrong. The exercise helped me connect a website name to a network address and compare information from different tools.

A publicly resolvable address is normal for a public website. I would treat it as part of the target profile, while keeping any further testing within the approved scope.

4.1.4 curl HTTP Response Headers

```
curl -I https://networkwalks.com
```

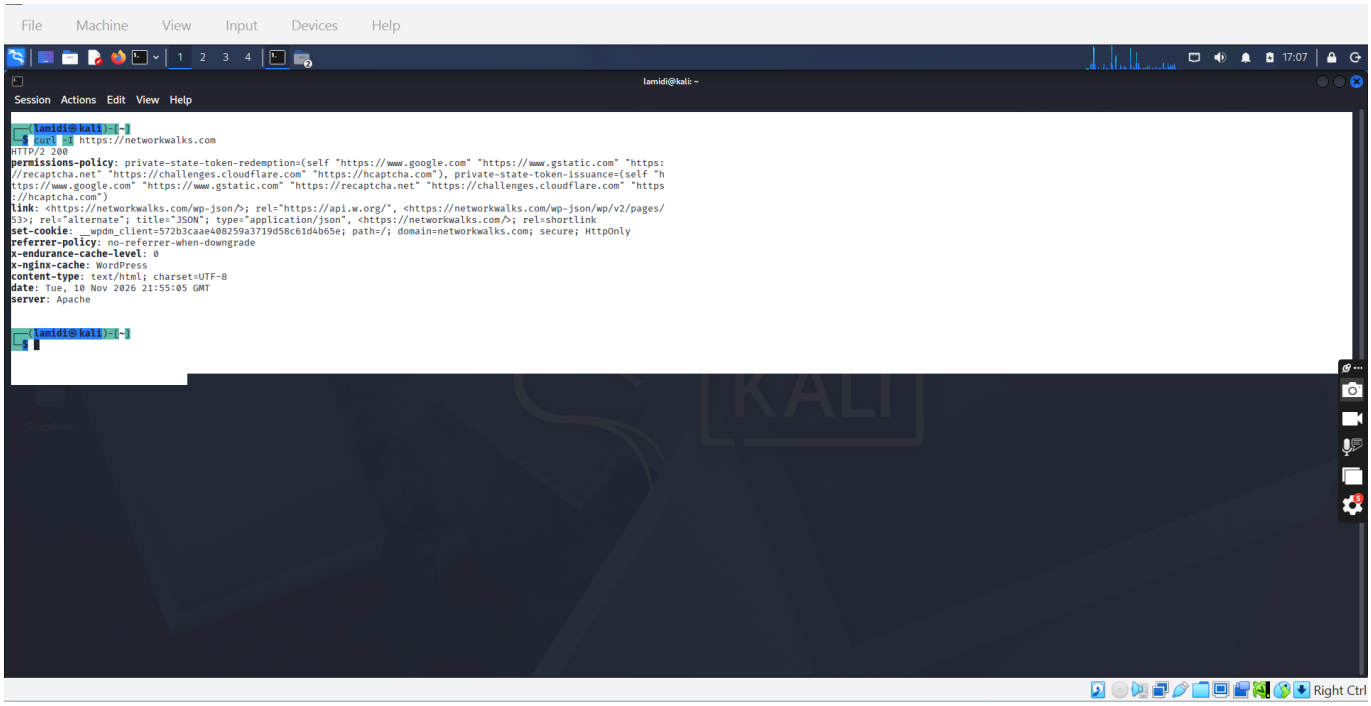


Figure 4 curl HEAD request and response headers | Evidence E05

I used curl with the -I option to view the response headers. The server returned HTTP/2 200. I observed server: Apache, content-type: text/html; charset=UTF-8 and x-nginx-cache: WordPress. The Link header included the WordPress REST API endpoint /wp-json/ and a reference to /wp-json/wp/v2/pages/53.

The response also included a cookie with Secure and HttpOnly attributes, a Permissions-Policy header and Referrer-Policy set to no-referrer-when-downgrade. The response date was 10 November 2026 at 21:55:05 GMT.

This exercise helped me understand the information available in HTTP headers. The REST API link indicates a WordPress feature; it does not by itself establish unauthorized access or a security flaw. I would review headers and exposed functionality against the site's requirements.

4.1.5 wafw00f Firewall Detection

```
wafw00f networkwalks.com
```

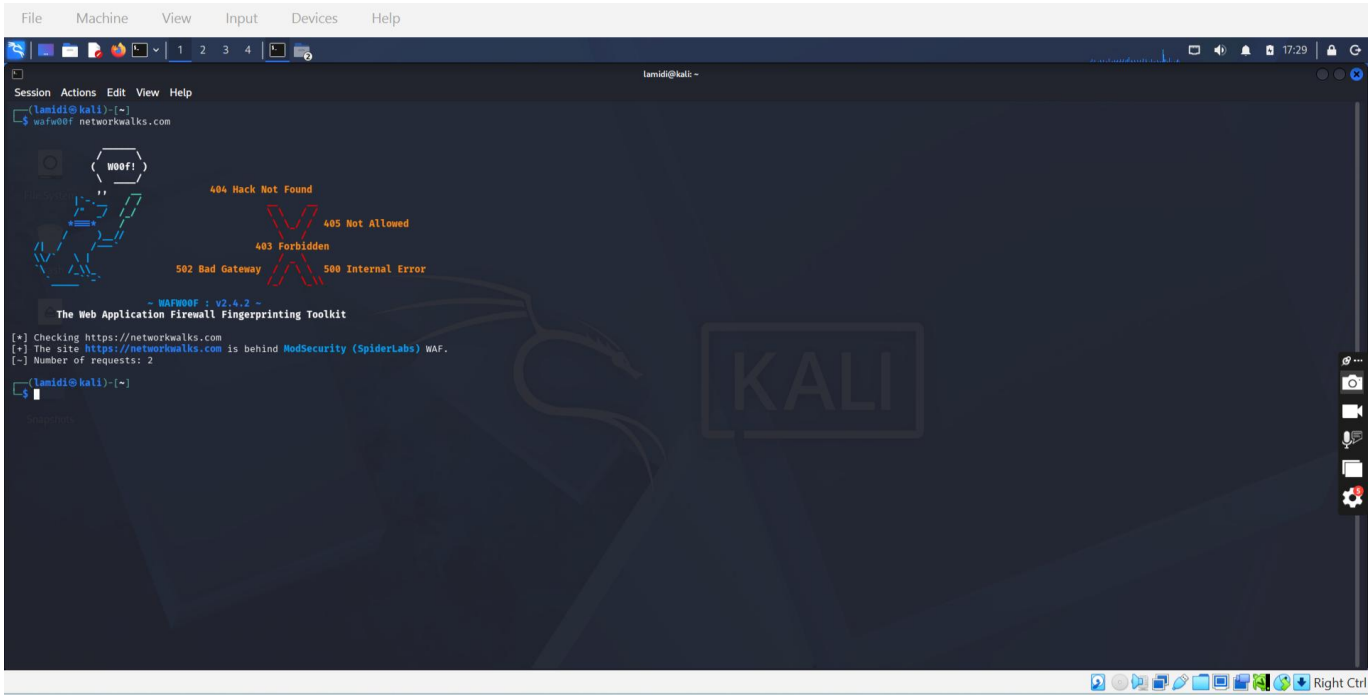


Figure 5 wafw00f reporting ModSecurity SpiderLabs | Evidence E06

I used wafw00f to check whether the website appeared to be protected by a web application firewall. Version 2.4.2 of the tool checked https://networkwalks.com and reported ModSecurity (SpiderLabs). The output showed that two requests were made.

I learned that a web application firewall can be part of a website's protection and that its presence may be identified from response behaviour. This result tells me what wafw00f detected during the check. It does not confirm the firewall's rule coverage, configuration or effectiveness.

My recommendation is to keep the WAF rules maintained and review its logs. Any assessment of whether it blocks specific attacks would require separate authorized validation.

4.1.6 DNSRecon DNS Enumeration

```
dnsrecon -d networkwalks.com
```

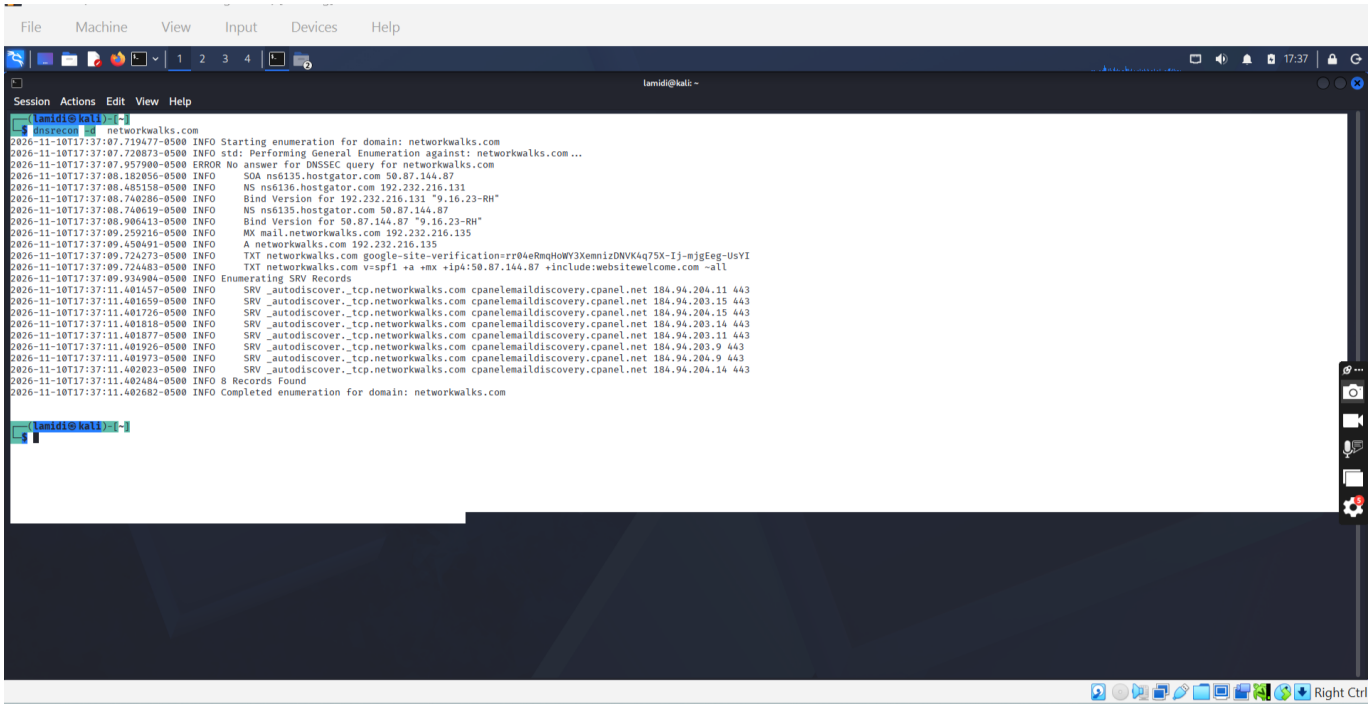


Figure 6 DNSRecon general enumeration results | Evidence E07

I used DNSRecon to review the DNS records associated with the domain. The A record returned 192.232.216.135. The output listed ns6135.hostgator.com at 50.87.144.87 and ns6136.hostgator.com at 192.232.216.131. It also showed mail.networkwalks.com at 192.232.216.135 and TXT records for site verification and SPF.

The SPF record ended in ~all, indicating a softfail policy. The output reported BIND version 9.16.23-RH for the listed DNS servers and eight autodiscover SRV results pointing to cpanelemaildiscovery.cpanel.net on port 443. DNSRecon also reported no answer for its DNSSEC query, consistent with the unsigned status shown by WHOIS.

I learned how DNS records reveal relationships between a website, name servers, mail services and service discovery. These records and the reported DNS software version need context before assigning risk. The provider may maintain the DNS service, so any changes should be coordinated with the responsible administrator.

4.2 Network Scanning with Zenmap

I used Zenmap to run a Ping scan on my own local /24 network. The generated command used nmap -sn for host discovery. The scan started on 11 November 2026 at 21:56 -0500 using Nmap 7.993. It scanned 256 IP addresses, found 10 hosts up and completed in 8.35 seconds.

I combined the scan output and topology in the figure below. Only 10.0.0.1, 10.0.0.8, 10.0.0.29 and 10.0.0.31 and their corresponding MAC addresses are shown. Other devices' addresses and identifying labels are obscured for privacy.

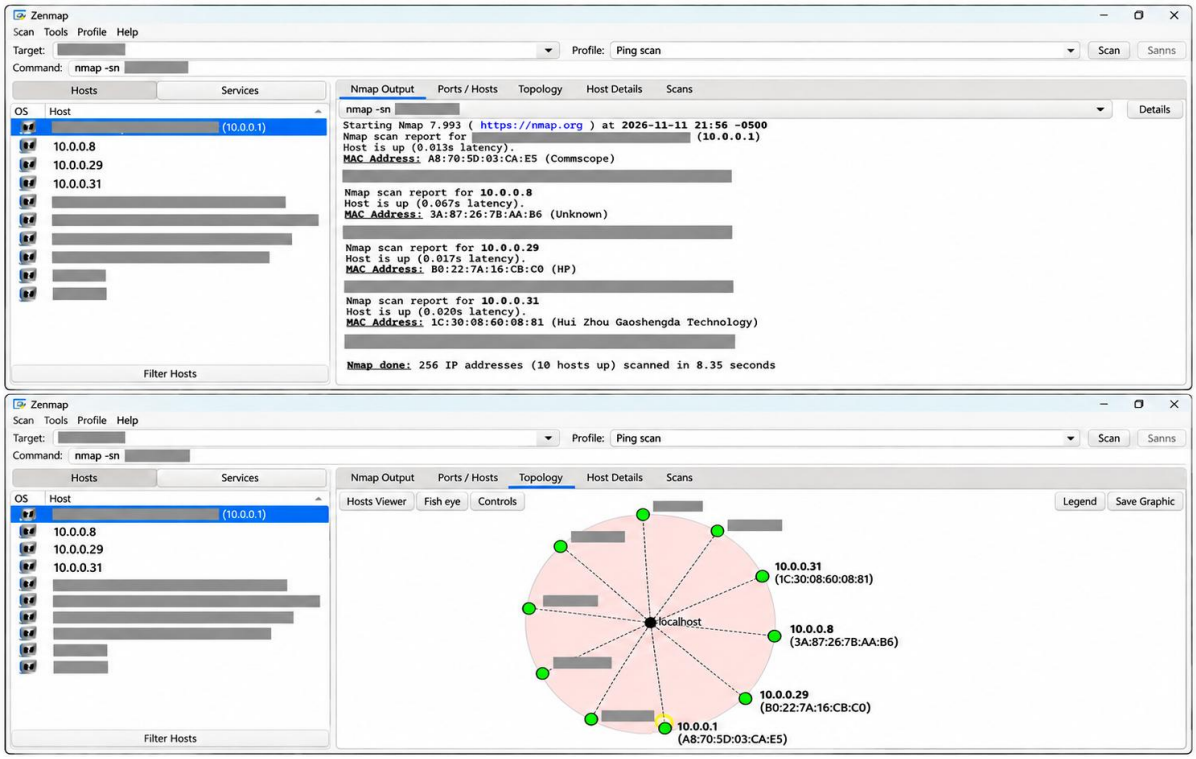


Figure 7 Zenmap scan output and annotated topology | Evidence E09 and E11

The upper panel shows the scan output. The lower panel shows ten green host nodes around localhost, with the four selected MAC addresses added as labels. This is an annotated view for documentation; the MAC labels are not a separate scan result. The topology illustrates discovered hosts and does not establish physical connections between the devices.

I learned how to compare scan output with a topology view, record selected devices and protect private details when sharing evidence. Host discovery does not establish open ports or vulnerabilities. The legend is not open in this image; a separate legend view, topology PDF export and Windows configuration evidence can be added if required.

4.2.1 Selected Device IP and MAC Addresses

I recorded the following four IP and MAC address pairs from the scan output. The other six devices' details remain hidden.

IP address	MAC address	Reported vendor
10.0.0.1	A8:70:5D:03:CA:E5	Commscope
10.0.0.8	3A:87:26:7B:AA:B6	Unknown
10.0.0.29	B0:22:7A:16:CB:C0	HP
10.0.0.31	1C:30:08:60:08:81	Hui Zhou Gaoshengda Technology

Table 1 Selected device inventory | Evidence E10

The vendor labels are those reported by Nmap. A MAC address or vendor label alone does not confirm the identity, ownership or security of a device. I would compare these results with my own approved device inventory.

5 Risk Analysis and Impact

I reviewed my observations to understand what information was visible and how it could support further assessment. The priorities below are preliminary review priorities, not validated vulnerability severity ratings. I did not perform exploitation or confirm any Critical vulnerability in these modules.

Observation	Evidence	Meaning and review priority
Website technologies visible	E03 WhatWeb	Medium review priority. Verify reported CMS and plugin versions, then check maintenance and applicable advisories.
Public IP address available	E03 and E04	Informational. The public address supports normal access and identifies the web service location.
HTTP headers and REST API links visible	E05 curl	Low review priority. Review unnecessary disclosure and confirm that public API access is appropriate.
ModSecurity detected	E06 wafw00f	Informational. The tool identified a security control; effectiveness was not tested.
DNS and mail information available	E02 and E07	Medium review priority. Review DNSSEC, SPF and reported DNS software with the responsible provider. Public records are not inherently vulnerabilities.
Ten hosts responded on my local network	E09 and E11	Informational. Host discovery and topology support inventory review. They do not establish unauthorized devices or vulnerabilities.

I understand that public information is often necessary for a website to function. My responsibility is to describe what I observed accurately, explain its possible relevance and avoid claiming a security weakness without sufficient evidence.

6 Recommendations

Keep the website software maintained. I recommend verifying the actual WordPress, plugin and library versions, applying supported updates and reviewing applicable security advisories.

Review website exposure. I recommend checking whether response headers reveal unnecessary details and confirming that REST API access matches the website's needs.

Review DNS and email configuration. I recommend confirming that public DNS records are required and accurate. The administrator should review DNSSEC and the SPF policy with the hosting or DNS provider.

Maintain and monitor the WAF. I recommend validating the ModSecurity deployment, keeping its rules current and reviewing logs for events that need attention.

Maintain a local device inventory. I recommend comparing authorized discovery results with a current list of approved devices and investigating unexpected entries.

Keep complete evidence and authorization. I will retain the commands, screenshots, testing scope and permission records. My Zenmap screenshots are included with four selected IP addresses visible and other identifiers obscured. I will retain any additional configuration and topology export evidence required for the submission.

7 Conclusion

This Week 2 practical helped me build confidence with Kali Linux and understand how different reconnaissance tools work together. I used WHOIS, WhatWeb, nslookup, curl, wafw00f and DNSRecon to collect and compare information about networkwalks.com.

I learned to explain what a command returned, support my observations with screenshots and separate findings from assumptions. My Zenmap scan and topology helped me understand device discovery and network documentation. I also learned to protect private network details when sharing evidence.

I am grateful to Networkwalks for this hands-on learning opportunity. The experience has strengthened my interest in cybersecurity and network security, and I look forward to developing my practical skills. I will continue to work within an authorized scope and report my results accurately.

8 Evidence Collected

My six footprinting screenshots are included in Section 4.1. My Zenmap scan and annotated topology are combined in Figure 7 in Section 4.2, with four selected IP and MAC pairs visible and the other devices' identifiers obscured. The register below connects each attachment to the relevant activity.

Evidence	Record	Location or status
E01	Written permission and testing scope	Permission recorded in the original report; document to be attached.
E02	WHOIS command and output	Figure 1 in Section 4.1.1.
E03	WhatWeb command and output	Figure 2 in Section 4.1.2.
E04	nslookup command and output	Figure 3 in Section 4.1.3.
E05	curl command and headers	Figure 4 in Section 4.1.4.
E06	wafw00f command and detection	Figure 5 in Section 4.1.5.
E07	DNSRecon command and records	Figure 6 in Section 4.1.6.
E08	Windows ipconfig output	To be attached for my actual local network.
E09	Zenmap command and scan results	Figure 7 upper panel in Section 4.2; four selected IP and MAC pairs visible.
E10	IP and MAC address inventory	Table 1 in Section 4.2.1 lists the four selected IP and MAC pairs.
E11	Zenmap topology and exported PDF	Figure 7 lower panel in Section 4.2. Legend view and PDF export can be added if required.

Author Attribution

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Github: Week-2--Penetration-Testing-Report/README.md at main · leleshin/Week-2--Penetration-Testing-Report

Project Information

Networkwalks Cybersecurity and Penetration Testing Internship
Week 02 | W2-PM1 and W2-PM5 | Repository GitHub